Project Report

Car Black Box (Event Data Recorder) Firmware

1. Executive Summary

The Car Black Box, or Event Data Recorder (EDR), is an advanced embedded system firmware developed for the PIC16 Microcontroller family. Functioning similarly to a commercial flight data recorder, this system is designed to continuously monitor vital vehicle parameters, including real-time speed, current gear, and accurate timestamps. When critical driving events occur, such as a collision or abrupt gear shift, the system permanently logs a data snapshot into an external EEPROM. Additionally, it features a highly secure, password-protected administrator dashboard for on-device log viewing, memory management, and data extraction via Serial UART.

2. Advanced Technical Features

- **Smart Memory Management (Circular Buffer):** To optimize the limited read/write cycles of the external I2C EEPROM, the system utilizes a circular buffer algorithm. Instead of halting when memory is full, the system dynamically recalculates memory address pointers to overwrite the oldest event with the newest event, ensuring continuous operation without memory overflow.
- **Modular Software Architecture:** The codebase is strictly modular. Hardware-specific drivers (ADC, UART, I2C, RTC, LCD, Matrix Keypad) are isolated into independent C source and header files, ensuring clean, scalable, and reusable code.
- **Non-Blocking State Machine Design:** The user interface and hardware switches are governed by a dynamic state machine. Hardware buttons intelligently change their behavior depending on whether the system is in "Driving Mode" or "Menu Navigation Mode," maximizing the utility of limited GPIO pins.
- **Hardware Timer Interrupts (ISR):** Utilizes microcontroller hardware timers and Interrupt Service Routines (ISRs) to handle background tasks like display blinking (cursor management) and polling without blocking the main execution loop.
- **Serial Data Extraction (UART):** Administrators can download the entire EEPROM event history formatted as a clean, readable table directly to a PC using UART at a 9600 baud rate.

3. Hardware & Architecture

The system is built on a bare-metal Embedded C foundation using the XC8 Compiler. It interfaces with various peripherals to achieve real-time monitoring and robust data logging.

Component / Protocol	Description
PIC Microcontroller	Core processing unit, handling state machines and peripheral communication.
I2C Protocol	Used to interface with the DS1307 Real-Time Clock and external EEPROM.
UART Protocol	Serial communication drivers for PC data transmission (9600 Baud).
ADC (Potentiometer)	Analog-to-Digital Converter used to simulate real-time vehicle speed.
16x2 Character LCD	Provides the visual dashboard for real-time monitoring and menu navigation.
Digital Matrix Keypad	Used for driving simulations (shifting, crashing) and menu interaction.

System File Structure

The firmware strictly separates concerns by isolating each peripheral into its own module. Key files include `main.c` (core logic), `i2c.c` (EEPROM/RTC comms), `uart.c` (Serial extraction), `ds1307.c` (Timekeeping), `clcd.c` (Display), `adc.c` (Speed simulation), and `isr.c` (Timer interrupts).

4. Hardware Interaction & UI State Machine

The system is built on a dynamic state machine where the digital keypad's behavior shifts entirely based on the current mode, reflecting advanced GPIO multiplexing.

State 1: Driving Dashboard Mode

When powered on, the system defaults to simulating a driving dashboard.

- **Potentiometer (ADC):** Acts as the throttle. Twisting it increases or decreases the simulated vehicle speed (0-99 km/h).
- **SW1 (Up)** / **SW2 (Down):** Shifts vehicle gears (e.g., Reverse → Neutral → Gear 1 → Gear 2). *Each shift triggers an EEPROM save.*

- **SW4 (Crash Event):** Simulates a critical collision (logged as 'C'). *Triggers an immediate EEPROM save.*
- **SW5 (Enter Menu):** Halts the dashboard and prompts the user for the 4-digit Admin PIN.

State 2: Administrator Menu Mode

Upon authenticating with the correct PIN, the switches adopt navigation capabilities:

- **SW1 & SW2:** Scroll up/down through menus or increment/decrement values in settings.
- **SW3:** Enter Sub-Menu or move the cursor to the next field (e.g., Hours to Minutes).
- **SW5:** Save and Confirm changes to the RTC or Password memory.
- **SW6:** Exit Menu and safely return to Driving Mode without saving.

5. Menu Capabilities & Serial Extraction

The protected Admin Menu exposes five core features designed for fleet managers or diagnostic engineers:

1. **View Log:** An interactive, scrollable list of the circular buffer's history, showing up to the 10 most recent driving events directly on the LCD.
2. **Clear Log:** Resets the EEPROM address pointers and clears the event count, effectively wiping the history.
3. **Download Log:** Transmits the formatted event history via UART to a PC.
4. **Set Time:** An interactive UI with a blinking cursor to manually configure the DS1307 RTC (Hours, Minutes, Seconds).
5. **Change Password:** Allows the administrator to update the secure 4-digit PIN.

Example UART Download Output

When a download is requested, data is formatted over serial as follows:

```
DOWNLOADING...
NO. TIME     EV  SP
0.  10:14:05 GR  00
1.  10:14:12 GN  00
2.  10:14:18 G1  15
3.  10:15:30 G2  34
4.  10:22:15 C   42
DOWNLOAD SUCCESS
```

6. Conclusion

The Car Black Box project is a comprehensive demonstration of advanced embedded systems engineering. By integrating multiple hardware protocols (I2C, UART), complex peripherals (RTC, ADC, Matrix Keypad), and intelligent memory management (Circular Buffer), the system successfully fulfills the robust requirements of a modern vehicle event data recorder. The modular software architecture ensures it is highly reliable and easily expandable for future automotive and IoT applications.